

Quantum-Classical Equivalence for AND Functions

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Sreejata Kishor Bhattacharya, Farzan Byramji, Arkadev Chattopadhyay, Yogesh Dahiya, Shachar Lovett

Tata Institute of Fundamental Research
University of California San Diego

Goal: Upper bound $\text{spar}(g)$ in terms of $\text{rank}^{1/3}(g \circ \text{AND}_2)$

- Assume, for the sake of contradiction, $\text{spar}(g)$ is large and $\text{rank}^{1/3}(g \circ \text{AND}_2)$ is small.

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- Assume, for the sake of contradiction, $\text{spar}(g)$ is large and $\text{rank}^{1/3}(g \circ \text{AND}_2)$ is small.
- From the second assumption, we have functions

$$a_i, b_i : \{0, 1\}^n \rightarrow [0, 1], i = 1, 2, \dots, m$$

and constants $\alpha_i, i = 1, 2, \dots, m$ such that

$$\left| g(x \wedge y) - \sum_{i=1}^m \alpha_i a_i(x) \wedge b_i(y) \right| \leq 1/3$$

- By standard techniques (eg John's lemma), we can assume $\sum |\alpha_i| \leq L$ (where $\log L \leq \text{polylog}(m)$)

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- We shall construct a function f (to be obtained from g) such that
 - Approximate degree of f is small
 - Degree of f is high

This contradicts known relations between degree and approximate degree (first proven by Nisan-Szegedy; constants were optimized later).

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This contradicts known relations between degree and approximate degree (first proven by Nisan-Szegedy; constants were optimized later).

- We shall get a low-degree approximating polynomial from the inequality

$$\left| g(x \wedge y) - \sum_{i=1}^m \alpha_i a_i(x) \wedge b_i(y) \right| \leq 1/3$$

whereas the high sparsity of g will guarantee the original polynomial has high degree.

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- Note that every function $f : \{0, 1\}^n \rightarrow \mathbb{R}$ can also be represented as a function $\tilde{f} : \{\pm 1\}^n \rightarrow \mathbb{R}$. It is easy to see that their degrees are the same.

Convention: For a function $f : \{0, 1\}^n \rightarrow \mathbb{R}$, we denote its equivalent function with the domain $\{\pm 1\}^n$ by $\tilde{f} : \{\pm 1\}^n \rightarrow \mathbb{R}$.

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- For upper-bounding the approximate degree, we shall look at functions in the $\{\pm 1\}^n$ domain. For lower bounding the degree, we shall look at functions in the $\{0, 1\}^n$ domain. Since the degree is invariant under change of domain, this is ok.

- **Observation:** Given the expression

$$g(x \wedge y) \approx \sum_{i=1}^m \alpha_i a_i(x) \wedge b_i(y),$$

we can hit each a_i by a restriction+averaging operator (ρ, δ) , replace each b_i by a constant, and get an approximation for g_ρ

Restriction+averaging operators:

A **restriction+averaging operator** is denoted by a string

$R \in \{0, 1, *, \Delta\}^n$. It converts a function $f : \{0, 1\}^n \rightarrow \mathbb{R}$ to a function $f : \{0, 1\}^{\text{free}(R)} \rightarrow \mathbb{R}$ where $\text{free}(R) = \{i \mid R_i = *\}$. The transformation is as follows:

- If $R_i = b \in \{0, 1\}$, fix the i -th coordinate to b .
- If $R_i = \Delta$, average over the i -th coordinate.
- If $R_i = *$, keep the i -th coordinate free.

We shall associate a partial assignment $\rho \in \{0, 1, *\}^n$ to each restriction+averaging operator R : ρ is obtained from R by changing all the Δ 's to 0's.

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- We shall define a distribution $\mathcal{D}$ over *restriction+averaging operators* R such that for any $\tilde{a} : \{\pm 1\}^n \rightarrow [0, 1]$,

$$\mathbb{E}_{R \sim \mathcal{D}}[|\tilde{a}_R|^{\geq d}] \leq \varepsilon$$

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- ... and such that $\mathbb{E}_{R \sim \mathcal{D}}[f_\rho] \geq D$, where ρ is the partial assignment associated with R .

- The first property, along with the expression for approximate rank, will give that $\text{approx-deg}(g_\rho) \leq d$ for a typical $R \sim \mathcal{D}$

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- The second property will give that $\text{deg}(g_\rho) \geq D$ for a typical $R \sim \mathcal{D}$.
- If the gap between d and D is large enough, we are done.

Obtaining an approximation for f_ρ

- Recall the expression we got from the low-approximate rank assumption:

$$g(x \wedge y) \approx \sum_{i=1}^m \alpha_i a_i(x) b_i(y)$$

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- Claim:** Given a restriction+averaging operator R , with ρ its associated partial assignment, there exist constants $c_1, c_2, \dots, c_m \in [0, 1]$ such that

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- Plugging $y_0 = (0, 1, 1, \dots, 1)$, $x = (1, z_2, z_3, \dots, z_n)$, we get

$$g(0, z_2, z_3, \dots, z_n) \approx \sum_{i=1}^m \alpha_i a_i(1, z_2, \dots, z_n) b_i(y_0)$$

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- Average these two expressions:

$$g(0, z_2, \dots, z_n) \approx \sum_{i=1}^m \alpha_i b_i(y_0) (a_i)_R(z)$$

Exhibiting a gap between degree and approximate degree

- **Conclusion:** If R is a restriction+averaging operator and ρ is the associated partial assignment, we get an approximation for g_ρ in terms of the $(a_i)_R$'s.

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- **Our next goal:** Using the fact that sparsity of g is high, we define a distribution $\mathcal{D}$ over restriction+averaging operators R such that:
 - For all $a : \{0, 1\}^n \rightarrow [0, 1]$, $E_{R \sim \mathcal{D}}[\tilde{\deg}(a_R)] \leq d$ (we will do something slightly different actually)
 - $E_{(R \sim \mathcal{D})}[\deg(g_\rho)] \geq D$ where ρ is the partial assignment corresponding to R .

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 - $E_{(R \sim \mathcal{D})}[\deg(g_\rho)] \geq D$ where ρ is the partial assignment corresponding to R .
- If we can do this with a sufficiently big gap between d and D , from the previous observation we get that the approximate degree of g_ρ is very small; whereas the second property implies the exact degree of g_ρ is very large. But Nisan-Szegedy implies for any Boolean function, degree and approximate degree are polynomially related. Contradiction.

Simplifying any bounded function

First requirement:

- This is what we will show:
For any $\tilde{a} : \{\pm 1\}^n \rightarrow [0, 1]$,

$$\mathbb{E}_{R \sim \mathcal{D}}[||\tilde{a}_R||^{\geq d}] \leq \varepsilon$$

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- If we achieve this, the total ℓ^1 Fourier mass above level d of the function

$$\sum_{i=1}^L \alpha_i c_i(a_i)_R$$

will be at most $L\varepsilon$, which we will set to be $< 1/100$. Thus, on expectation, the approximate degree of g_ρ will be $\leq d$.

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Why switching to the $\{\pm 1\}$ basis is helpful:

- Let $\tilde{f} : \{\pm 1\}^n \rightarrow \mathbb{R}$ be any function. Its Fourier expansion is

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- Thus, averaging out over a variable kills all monomials containing it.

Defining the distribution

- Assume $g : \{0, 1\}^n \rightarrow \{0, 1\}$ has $\geq s$ non-zero monomials in the deMorgan basis. Let $\mathcal{F} \subseteq 2^{[n]}$ denote the set of nonzero monomials.

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$$R(i) = \begin{cases} * & \text{with probability } p \\ \Delta & \text{with probability } 1 - p \end{cases}$$

(Here p is small but non-negligible.)

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- Let $A = \{i | i \in T, R(i) = *\}$. Look at the polynomial

$$g_{A,T} = \sum_{S: S \cap T = A} \alpha_S x_{S \setminus A}$$

This is defined on $\{0, 1\}^{[n] \setminus T}$, and is nonzero since T is shattered. So there exists a $y \in \{0, 1\}^{[n] \setminus T}$ such that $g_{A,T}(y) \neq 0$.

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- Set $R(i) = y_i$ for $i \notin T$.